

Final Project Report

Water Supply Interruption Events in the City of Split

Project Idea

The initial idea was to develop a simple geospatial application for monitoring water supply interruptions in the City of Split. The main goal was to collect interruption notices from ViK Split, geocode the affected locations, and visualize them together with existing water infrastructure.

The original plan also considered using infrastructure data from the Infrastructure Cadastre (SKI). However, this option was abandoned because the infrastructure data was not easily accessible for direct use in the project. The focus was therefore shifted to publicly available data from ViK and OpenStreetMap.

The final goal was to create a workflow that could show where and why water supply interruptions occur in Split, while also exploring the possibility of automating the data collection.

Data Sources

Two main data sources were used for the project. ViK Split website provided the information about water supply interruptions (*Figure 1.*) Vodovod i kanalizacija d.o.o. Split (ViK Split) is public utility provider managing water supply and wastewater infrastructure in Split, Croatia. Data collected includes date, affected location, cause, start time, end time and source URL. The final dataset containing 43 interruption records was saved in the *vik_split_city_interruption_dataset.xlsx*.

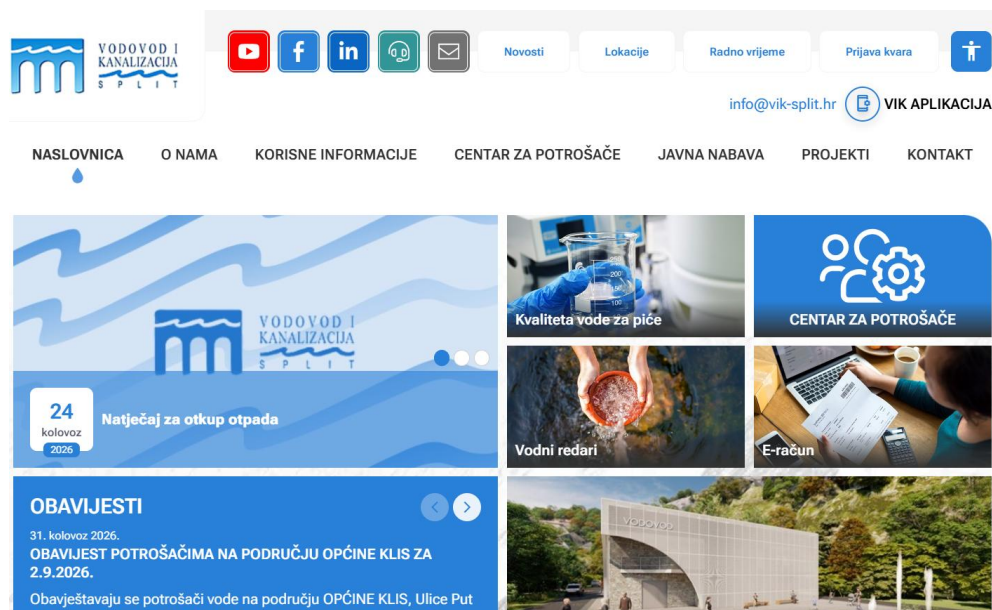


Figure 1. ViK Split

OpenStreetMap (OSM) was used to obtain water-related infrastructure features and the spatial boundary of the City of Split. Nominatim, the geocoding service used by OSM, was tested for converting the reported street locations into geographic coordinates (*Figure 2*).

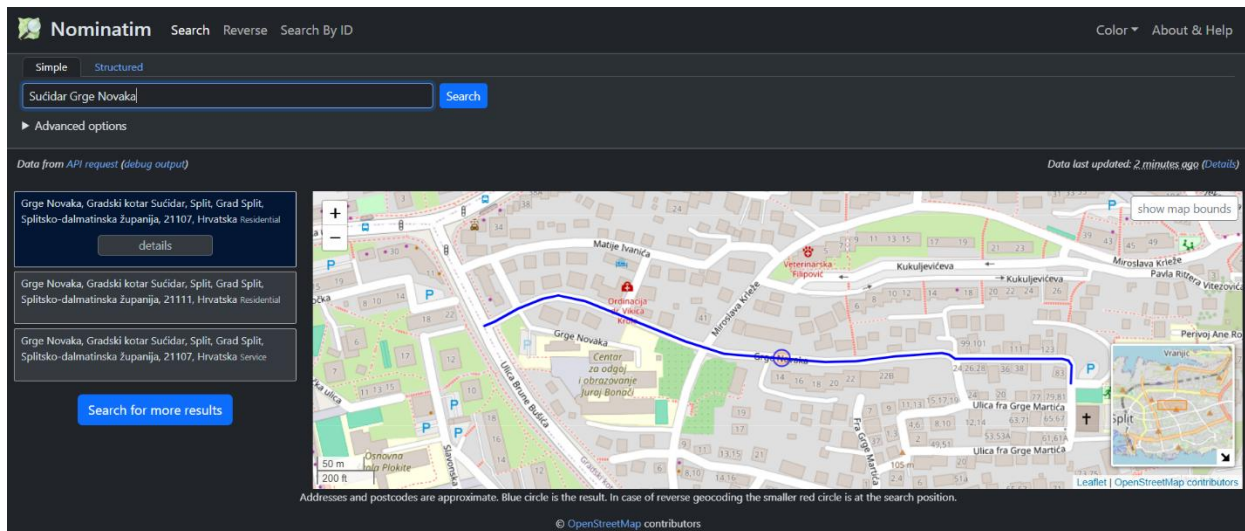


Figure 2. Nominatim

Methodology

The workflow was implemented in Python using Pandas, GeoPandas, OSMnx and Matplotlib.

First, the ViK interruption dataset was loaded and cleaned. The reported locations were then geocoded using Nominatim. Since automatic geocoding did not consistently produce reliable results for all locations, the final coordinates were manually checked and added where necessary.

The interruption points and OSM water infrastructure were converted into spatial datasets and projected to a metric coordinate system. The City of Split boundary was used to limit the analysis area.

Finally, the data was analysed by year, cause, duration and location. The results were exported as GeoJSON and CSV files, while a final map was created to visualize the interruption points, water infrastructure and the City of Split boundary.

Results

The final workflow successfully converted the ViK interruption records into a spatial dataset. All 43 interruption records were geocoded, allowing them to be displayed on a map together with the OSM water infrastructure.

The analysis was also extended beyond simply counting the mapped points. The records were grouped by year, cause, duration and location, and the results were exported into separate Excel sheets for further inspection:

- Overview - a total of 43 records, 43 successfully geocoded records, 4 years, 9 different causes, and the most common cause, *water supply network works*, accounting for 33 events (76.7%).
- Causes - analysis of interruptions by cause.
- Causes by Year - analysis of interruption causes by year.
- By Year - analysis of the number of interruptions by year.
- Locations - analysis of the locations where interruptions were recorded.

The recorded interruptions were mainly associated with two types of causes: planned network works and faults/repairs. The map shows that interruptions are distributed across different parts of Split, with several events appearing relatively close to each other in the central parts of the city (*Figure 3.*).

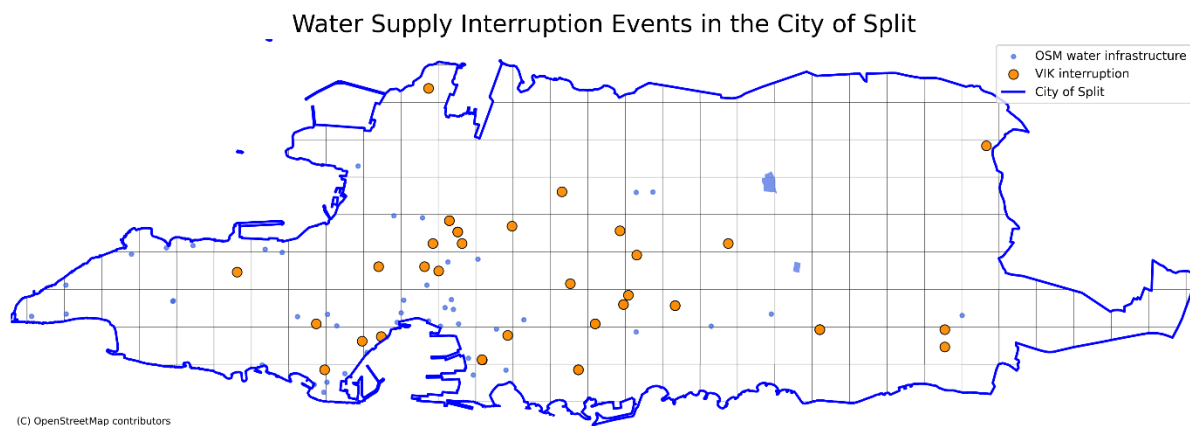


Figure 3. Map of the results

Results are gathered in several output files:

vik_split_interruption_map.png - final map showing the recorded water supply interruption locations within the City of Split.

vik_split_interruption_events.geojson - spatial dataset containing the 43 geocoded water supply interruption events as point features.

vik_split_water_infrastructure.geojson - spatial dataset containing 48 OSM water infrastructure features downloaded for the City of Split.

vik_split_summary.csv - summary table reporting 43 total interruption records, 43 geocoded records, a 100% geocoding rate, and 48 OSM water infrastructure features

vik_split_interruption_analysis.xlsx - detailed Excel analysis containing an overview of the 43 interruption records, analyses by cause and year, the distribution of interruption causes across years, and the recorded interruption locations

Discussion & Outlook

The final result is useful for a basic spatial overview, but several problems appeared during the development of the workflow. The biggest issue was data availability. The initial idea was to use the Infrastructure Cadastre (SKI) as the main source for water infrastructure (*Figure 4.*). However, the data was not straightforward to access and use for this project, so this approach had to be abandoned. OpenStreetMap was used instead. Although OSM was easy to access through OSMnx, it does not provide a complete representation of the actual water supply network. In the final download, 48 water-related OSM features were found, but no actual pipeline segments were available for the study area. This means that the OSM infrastructure layer should only be considered as a reference layer and not as a complete map of the ViK water network.



Figure 4. Infrastructure Cadastre (SKI)

Another major problem was the collection of interruption data. The original plan was to automatically scrape interruption notices from the ViK Split website. This did not work because the website's robots.txt restricts automated access to the relevant archive (Figure 5.). Because bypassing this restriction was not appropriate, the interruption records had to be collected manually. This also meant that the final workflow is not fully automated. Instead of continuously collecting new notices, it currently analyses a manually prepared dataset. This is an important difference from the original project idea.

```
[1/5] Checking robots.txt...
robots.txt does NOT allow automated access to the archive.
Stopping here. We will not bypass the restriction.
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Figure 5. Restricted access

The geocoding process also required several iterations. Nominatim was able to understand Croatian characters such as č, ć, ž, š and đ and could successfully locate some Split streets. However, complex descriptions were much harder to process. Many ViK notices contain several streets, intersections, neighbourhoods or descriptions of an entire area instead of one specific address. Nominatim could not reliably interpret every location description as one geographic point. Different versions of the geocoding workflow produced different results, including incorrectly located features. For this reason, the final 43 records were manually checked and corrected to make sure that all interruption locations could be represented spatially.

The spatial analysis itself was also affected by the quality of the input data. Because the dataset was relatively small and manually collected, the results should not be interpreted as a complete statistical picture of water supply interruptions in Split. The 500 m grid was therefore used mainly to visualize the spatial distribution of recorded events and identify areas where events appear to be concentrated. It should not be interpreted as proof of statistically significant hotspots.

There were also several technical problems during map production. The OSM basemap could not be included, same as CARTO basemap which require API key (Figure 6.). The final map still contains the City of Split boundary, interruption locations, OSM water-related features and the grid, but without the background street map.

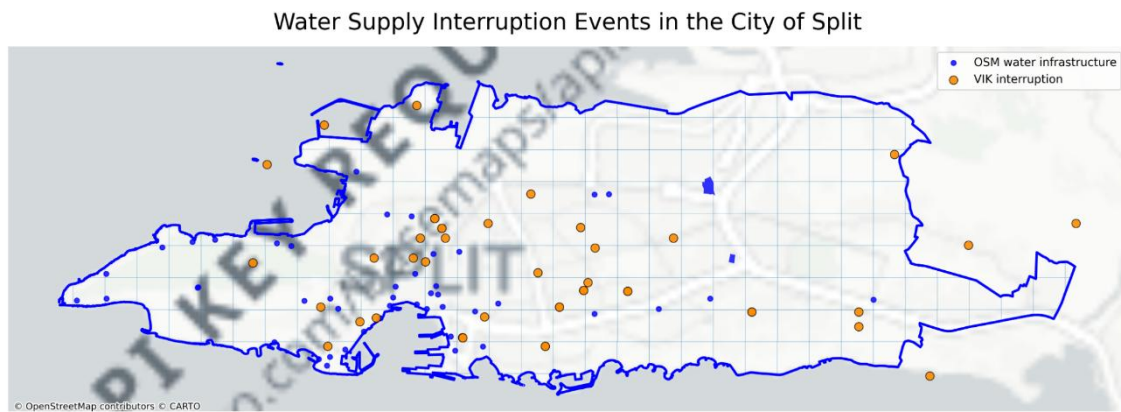


Figure 6. Basemap troubleshooting

Finally, the analysis output changed several times during development. The Excel summary initially focused mainly on the number of successfully geocoded features, which was not very useful for answering the actual research question. It was therefore changed to include information about interruption causes, years, locations and other attributes. This reflects one of the main limitations of the project: the final workflow developed through several iterations because some of the original assumptions about the available data and the feasibility of automation were not realistic.

Overall, the project became considerably more complicated than I expected in the beginning, and I am not fully satisfied with the final result. I found myself getting lost quite a lot during the development process, especially as several technical and data-related problems appeared at the same time. The main challenge was not the spatial analysis itself, but obtaining reliable and automatically accessible data. Although the workflow can successfully transform interruption records into a spatial dataset and produce a basic analysis, it does not yet function as the fully automated monitoring application I envisioned. My key takeaway from this project is how different an initial project idea can be from its actual implementation, especially when working with real-world data and unexpected technical limitations. For future projects, I would focus more on defining a realistic scope from the beginning, while also improving my troubleshooting and problem-solving approach when things do not work as expected.

References

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AI Declaration

I, Andjela Baković, hereby declare that artificial intelligence (AI) was used within this project. OpenAI ChatGPT was used as a supporting tool during the development of this project to:

- clarify Python concepts and explain parts of the code
- assist with troubleshooting and debugging errors
- suggest possible solutions when technical problems occurred
- improve the wording and grammar of the final report

The project idea, data collection, implementation, analysis and final decisions were carried out by the author.